

Charulkumar Chodvadiya

✉ charulpatel2499@gmail.com —  /charulp2499 —  /charulp2499 —  Google Scholar —  Portfolio

AI/ML with 3+ years of experience specializing in large-scale model pre-training, parameter-efficient fine-tuning of LLMs/VLMs, and edge-optimized AI deployment. Proven track record of translating complex AI research into scalable, low-latency production pipelines. Expertise spans from-scratch distributed training on large-scale datasets to inference optimization using quantization, knowledge distillation, and pruning for high-throughput AI APIs.

Skills

Programming:	Python (Primary), C++, SQL
AI/ML:	Machine Learning, Deep Learning, Generative AI, Multi-modal AI, Document Intelligence, Speech AI, Time Series & Statistical Analysis
Architectures:	Classical ML Models, CNNs, ResNet, YOLO, U-Net, RNNs, LSTMs, Autoencoders, Transformers, Vision Transformers (ViTs), GANs, Diffusion Models
Model & Distributed Training:	Distributed Data Parallel (DDP), Fully Sharded Data Parallel (FSDP), Multi-GPU Training, Gradient Accumulation, Gradient Checkpointing, Mixed-Precision Training
Model Optimization:	Quantization, Knowledge Distillation, Pruning, TensorRT, ONNX Runtime, CUDA Optimization
Gen AI:	Large Language Models (LLMs), Vision-Language Models (VLMs), Retrieval-Augmented Generation (RAG), Multi-Agent Systems, Prompt Engineering, RLHF/RLAIF, PEFT (LoRA, QLoRA)
Computer Vision:	Medical & Multi-Spectral Imaging, Image Processing, Object Detection, Image Segmentation, Depth Estimation, Anomaly Detection, Contrastive Learning
Frameworks & Libraries:	PyTorch, TensorFlow, Hugging Face, OpenCV, Detectron2, MMDetection, LangChain, LangGraph, NumPy, Pandas, Scikit-learn, SciPy, XGBoost, Matplotlib, librosa
MLOps & Deployment:	FastAPI, Docker, AWS (EC2, Lambda, Bedrock), Kubernetes, MLflow, vLLM, Weights & Biases (W&B), Git, Linux, CI/CD
Databases:	PostgreSQL, MongoDB, Pinecone, Weaviate

Experience

AI Engineer

Radin Health

Sep 2025 – Present

Remote, USA

- Trained vision models (U-Net, YOLO, and Custom CNNs) for medical imaging tasks using CT/MRI datasets; engineered distributed multi-GPU training pipelines (DDP, FSDP) with gradient accumulation and checkpointing to scale large workloads.
- Fine-tuned and deployed Vision-Language Models (VLMs) for medical document intelligence, achieving 96–99% metadata extraction accuracy across 8,000+ multi-page healthcare documents daily; built scalable asynchronous APIs with background task offloading to minimize user-perceived latency.
- Built token-optimized LLM pipelines processing 20,000+ clinical records daily for automated QA, MIPS compliance, CPT/RVU and ICD-10 extraction, and AI-generated follow-up recommendations; reduced inference costs by 30–40% through dynamic prompting and context compression while maintaining 96–99% accuracy.
- Developed real-time patient-facing AI voice bots and conversational chatbots, integrating WebSocket-based STT/TTS pipelines with 10–15 ms streaming latency and 92% end-to-end task success; fine-tuned domain-specific speech recognition models, reducing transcription and reporting errors by ~80%.
- Applied quantization, knowledge distillation, and pruning with optimized inference runtimes to improve model efficiency and reduce computational requirements for production deployment.
- Architected an asynchronous multi-agent AI copilot automating end-to-end clinical front-desk operations, spanning document verification, call scheduling, pending-action resolution, and patient follow-ups.

AI Developer

HubBroker ApS

Jan 2025 – Aug 2025

Ahmedabad, India

- Fine-tuned and deployed LoRA-based LLMs and VLMs for multilingual European business document intelligence, achieving 94% extraction accuracy on structured, semi-structured, and tabular documents.
- Developed AI-driven document quality classification systems achieving 99% accuracy across diverse business document types, enabling automated validation and processing workflows at scale.
- Designed and deployed production-grade FastAPI services integrating LLM/VLM inference into enterprise automation workflows, with GPU-optimized inference and scalable deployment pipelines.

AI Research Intern

Indian Institute of Technology Hyderabad (VIGIL Lab)

Jun 2023 – Apr 2024

Hyderabad, India

- Researched federated learning for brain-tumor segmentation across decentralized medical imaging datasets, training and evaluating U-Net, Masked U-Net, and ResNet architectures with each federated client running on a dedicated GPU.
- Developed uncertainty-aware U-Net variants and a contrastive-learning-based global-local objective to improve segmentation robustness under heterogeneous federated clients.
- Achieved a 5–12% improvement in segmentation performance over baseline federated training approaches across evaluated experimental settings.

Research Collaboration

AI Researcher

2023 – 2024

Koita Center of Digital Health (IITB)

- Conducted research in medical imaging, developing U-Net-based segmentation models and combining supervised objectives with contrastive learning to improve robust feature representation and segmentation performance.
- Developed a novel Feature-Enhanced Spatial Segmentation (FeSS) loss function, improving segmentation accuracy by 8–11% and resulting in a first-author publication at IEEE ISBI 2024.
- Researched EEG signals using classical machine learning and time-series modeling approaches for physiological signal analysis and classification.

Education

Masters in Technology - Artificial Intelligence

Pandit Deendayal Energy University

(Gold Medalist) (CGPA: 8.94 / 10.00)

Aug 2022 – May 2024

- Relevant Coursework: Computer Vision, Deep Neural Networks, Machine Learning, Time Series Data Analysis, Pattern Recognition and Machine Learning, Explainable AI, NLP, Cloud Infrastructure for AI
- **Teaching Assistant (Research):** Conducted research in machine learning, statistical data modeling, and computer vision. Key projects included solar power forecasting and maintenance prediction, night vision face detection, and palm print identification.
- IEEE CIS Student Branch Chapter Chair / IEEE CS Student Branch Chapter Vice-chair

Bachelor of Engineering - Computer Engineering

Gujarat Technological University

(First Class with Distinction) (CPI: 7.95 / 10.00)

Jul 2017 – Jun 2021

Selected Projects

Medical Image Segmentation — Centralized & Federated

[Code \(Centralized\)](#) | [Code \(Federated\)](#)

- Developed 3D brain tumor segmentation pipelines on multi-institutional MRI datasets in both centralized and federated learning settings; proposed a novel Feature-Enhanced Spatial Segmentation (FeSS) loss function that improved segmentation accuracy by 8–11%, resulting in a publication at IEEE ISBI 2024.
- Implemented federated learning across decentralized medical imaging clients using a contrastive learning-based local training objective, mitigating data heterogeneity and improving segmentation performance by 5–12%.

SynthAI: Interactive Generative Image Sandbox

[Live Space](#)

- Built an interactive web platform on Hugging Face Spaces enabling real-time image generation and side-by-side comparison across multiple GAN architectures, designed for research exploration and generative model benchmarking.

Real-Time Depth Estimation & 3D Point Cloud Reconstruction

[Code](#) | [Demo](#)

- Developed a real-time depth estimation system combining MiDaS depth prediction and YOLOv5 object detection on live webcam streams, enabling 3D scene understanding and point cloud reconstruction for spatial perception applications..

PalmScanX: Contactless Palmprint Biometric Authentication

[Code](#)

- Created a proprietary palmprint dataset comprising 3,600+ annotated images and developed contactless biometric authentication pipelines using YOLOv5 and YOLOv8, incorporating image enhancement, noise reduction, and transfer learning techniques for robust palmprint detection.

Publications

- [C. Chodvadiya](#), N Mahla, KG Singh, and KS Jadhav, "FESS Loss: Feature-Enhanced Spatial Segmentation Loss for Optimizing Medical Image Analysis," *2024 IEEE International Symposium on Biomedical Imaging (ISBI)*. [\[Link\]](#)
- [C. Chodvadiya](#), V. Solanki, and KG Singh, "Intelligent Virtual Worlds: A Survey of the Role of AI in the Metaverse," *2024 3rd International Conference for Innovation in Technology (INOCON)*. [\[Link\]](#)
- SK Gaurav, [C. Chodvadiya](#), B Chintan, S Pooja, and A Bruno, "Seeing in the Dark: A Different Approach to Night Vision Face Detection with Thermal IR Images," (2023) CEUR-WS under AIXIA. [\[Best Practical Paper\]](#) [\[Link\]](#)
- [C. Chodvadiya](#), KG Singh, and R Vyas, "Detection of palmprint from contactless smartphone-based video hand dataset." NCVPRIPG 2023. Singapore: Springer Nature Singapore, 2023. [\[Link\]](#)
- KG Singh, [C. Chodvadiya](#), S Vyas, H Patel, S Mishra, and B Tripathi, "Prediction of Solar Power Generation and Maintenance Activities for 1MW Power Plant," *2023 IEEE IAS Global Conference on Emerging Technologies (GlobConET)*, 2023. [\[Link\]](#)